

Toward Generalized Web Agent Training: A Deep Dive into Entropy-Balanced Reinforcement Learning

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Abstract

Recently, Agentic Reinforcement Learning (Agentic RL) has made significant progress in incentivizing the multi-turn, long-horizon tool-use capabilities of web agents. While mainstream agentic RL algorithms autonomously explore high-uncertainty tool-call steps under the guidance of entropy, excessive reliance on such signals leads to the training collapse. In this paper, we delve into the challenges caused by entropy and propose the Agentic Entropy-Balanced Policy Optimization (AEPO), an agentic RL algorithm designed to balance entropy in both the rollout and policy update phases. AEPO comprises two core components: (1) a dynamic entropy-balanced rollout mechanism that adaptively allocates global and branch sampling budget through entropy pre-monitoring, while imposing a branch penalty on consecutive high-entropy tool-call steps to prevent over-branching issues; and (2) Entropy-Balanced Policy Optimization that inserts a stop-gradient operation into the high-entropy clipping term to preserve and properly rescale gradients on high-entropy tokens, while incorporating entropy-aware advantage estimation to prioritize learning on high-uncertainty tokens. Results across 14 challenging datasets show that AEPO consistently outperforms 7 mainstream RL algorithms. Further analysis reveals that AEPO improves rollout sampling diversity while maintaining stable policy entropy, facilitating scalable training of general web agents. The code is available at <https://github.com/RUC-NLPIR/ARPO>.

CCS Concepts

• Information systems → Web applications; • Computing methodologies → Reinforcement learning.

Keywords

Agentic Reinforcement Learning, Agentic Search, Tool Learning

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1 Introduction

The emergence of large language models (LLMs) has profoundly revolutionized a wide range of reasoning tasks [18, 41, 52, 64]. Despite their impressive capabilities, the static nature of their internal knowledge often leads LLMs to experience hallucinations in knowledge-intensive scenarios [68]. Retrieval-augmented generation (RAG) empowers LLMs to reason with retrieved relevant knowledge, improving the reliability of generated answers [11, 19, 30]. However, with the explosive growth of web information, the static RAG workflow limits effective interaction between LLMs and search engines, resulting in bottlenecks in open-domain web exploration. To overcome these challenges, a series of LLM-based web agents have emerged [33, 35]. These agents perform on-demand web searches during reasoning and strategically interact with external environments, achieving in-depth web information seeking [26, 32].

To strive for efficient training of web agents, early implementations focus on distilling tool-use trajectories from stronger models and guide weaker models through supervised fine-tuning (SFT) [9, 17, 31]. However, relying solely on SFT struggles to discover autonomous and generalizable tool-use capability [6]. As large-scale reinforcement learning with verifiable rewards (RLVR) demonstrates the potential to unlock frontier LLM capabilities [18, 51, 53], several web-search agents adopt trajectory-level RL [45, 66] combined with carefully designed reward functions to elicit agentic reasoning in LLMs [8, 34, 59]. While effective to some extent, this line of work consistently overlooks the multi-turn interactive nature between LLMs and tool environments [67], making it difficult to discover step-level tool-use behaviors during RL training. To mitigate this limitation, recent efforts in agentic RL have shown that web agents often display high entropy in their output tokens due to uncertainty about the external tool-call results. Drawing

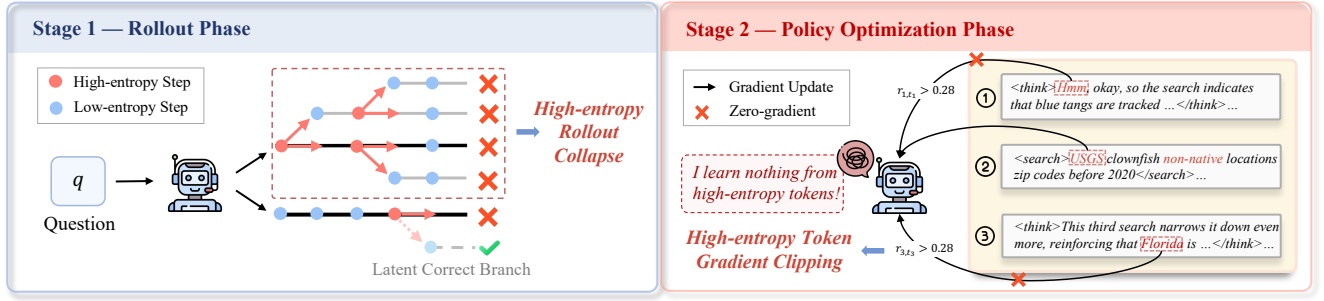


Figure 1: Two high-entropy challenges in agentic RL. (1) High-Entropy Rollout Collapse: Over-branching at high-entropy steps along specific paths, limiting exploration of other potential correct branches; (2) High-Entropy Token Gradient Clipping: Consistent clipping of high-entropy token gradients during policy updates hinders learning effective exploration behaviors.

on this finding, they introduce a tree-structured rollout method that adaptively branches at high-entropy tool-call steps, effectively broadening sampling diversity and coverage [10, 14, 24].

Although these entropy-driven agentic RL algorithms stimulate exploration of latent tool-use behaviors, such high-entropy signals further pose extra challenges for web agent training:

- (1) **High-entropy Rollout Collapse:** During the rollout phase, high-entropy tool-call steps often occur consecutively, leading LLMs to over-branch along a single trajectory under high-entropy guidance. This situation depletes the branching budget for other trajectories at high-entropy steps, ultimately limiting the diversity and scope of rollout sampling (Figure 1 (left)).
- (2) **High-entropy Token Gradient Clipping:** The tree-structured rollout strategy encourages LLMs to explore step-level tool-use behaviors, thus preserving valuable high-entropy tokens. However, vanilla RL algorithms aggressively clip high-entropy tokens' gradient during the policy update, leading to the premature termination of the LLM's exploration (Figure 1 (right)).

To address these challenges, we propose **Agentic Entropy-Balanced Policy Optimization (AEPO)**, an entropy-balanced agentic RL algorithm designed for training multi-turn web agents. Unlike traditional entropy-driven RL approaches [10, 36], AEPO focuses on balancing and rationalizing rollout branching and policy updates under the guidance of high-entropy tool calls, thereby achieving more stable RL training. Specifically, we pioneer the quantification of two inherent entropy-driven challenges on agentic RL.

Building on these insights, AEPO introduces two key algorithmic optimizations: (1) **Dynamic Entropy-balanced Rollout Mechanism:** To mitigate "High-entropy Rollout Collapse" issue, AEPO initially proposes the entropy pre-monitoring to adaptively allocate global and branch sampling budget, ensuring balanced exploration across the tree-structured rollout. Moreover, it incorporates a branch penalty strategy for consecutive high-entropy tool-call steps to effectively address over-branching issues in specific chains. (2) **Entropy-Balanced Policy Optimization:** To tackle the "High-Entropy Token Gradient Clipping", we intuitively integrate a *stop-gradient* operation into the high-entropy clipping term during policy updates. This preserves and properly rescales gradients of high-entropy tokens during backpropagation while leaving the forward pass unchanged. Furthermore, AEPO proposes entropy-aware

advantage estimation, integrating entropy advantage to prioritize learning on high-uncertainty tokens.

We conduct comprehensive evaluations across 14 datasets covering deep information seeking, knowledge-intensive and computational reasoning. The results show that AEPO consistently outperforms seven mainstream RL algorithms. In summary, the key contributions are as follows:

- We systematically reveal two entropy-driven issues inherent to agentic RL: "High-Entropy Rollout Collapse" and "High-Entropy Token Gradient Clipping". Through preliminary experiments, we quantify their impact on multi-turn web-agent training, offering empirical evidence for further research into entropy balancing.
- We propose a **Dynamic Entropy-Balanced Rollout** mechanism, which adaptively allocates rollout sampling budgets via entropy pre-monitoring, while imposing a branch penalty on consecutive high-entropy steps to prevent over-branching issues.
- We introduce **Entropy-Balanced Policy Optimization**, which intuitively integrates a *stop-gradient* operation into the high-entropy clipping term to preserve and rescale gradients on high-entropy tokens, while incorporating entropy-aware advantage estimation to prioritize learning on high-uncertainty tokens.
- Experiments on 14 challenging benchmarks demonstrate that AEPO consistently outperforms mainstream RL algorithms in web agent training. Quantitative analyses across *rollout diversity*, *tool-call efficiency*, and *entropy dynamics* verify AEPO's strong scalability and stability for training general web agents.

2 Related Work

2.1 Reinforcement Learning for Web Agent.

Recent advances in agentic reinforcement learning (RL) [67] have accelerated efforts to train general-purpose web agents. Early systems [3, 47] demonstrated autonomous search-engine interaction with rule-based RL, inspiring subsequent work on tool-augmented and reward-improved training, such as R1-Searcher++ [48] and Toolstar [8]. In parallel, several studies [16, 25, 63] investigate asynchronous training pipelines to scale web-agent learning. Building on the post-training paradigm, Websailor [49, 62] incorporates data synthesis, RL optimization and report generation. More recently, algorithmic improvements for web agents [10, 15, 36] adopt tree-structured rollouts to enable high-entropy branch sampling. Despite

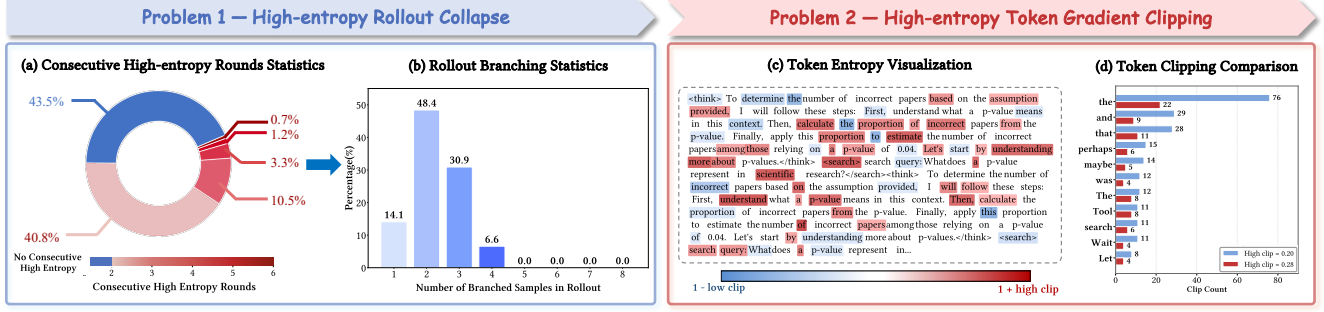


Figure 2: Quantitative statistics of two entropy-based challenges in web agent RL training.

these advances, existing methods largely overlook challenges introduced by high-entropy tokens; we address this gap with AEPO.

3 Preliminary

Before introducing the AEPO algorithm, we will briefly outline key task definitions and illustrate preliminary entropy-based experiments to reveal key limitations of web agent RL training.

3.1 Problem Definition

3.1.1 Agentic Reinforcement Learning. In this section, we define the training objective for agentic reinforcement learning as follows:

$$\max_{\pi_{\theta}} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta}(\cdot | x; T)} [r_{\phi}(x, y)] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(y | x; T) \| \pi_{\text{ref}}(y | x; T)],$$

Here, T is the available tool set, π_{θ} and π_{ref} denote the policy and the reference LLM. The symbols r_{ϕ} is the reward function. The input x is drawn from the dataset $\mathcal{D}$ and y is the output.

3.1.2 Token Entropy Calculation. Suppose V is the size of the vocabulary, $\mathbf{z}_t \in \mathbb{R}^V$ are the logits before applying the softmax function, and τ is the temperature parameter for decoding, inspired by recent studies in entropy-based RL efforts [5, 57, 58, 69], we determine the entropy of token generation at step t using the formula:

$$H_t = - \sum_{j=1}^V p_{t,j} \log p_{t,j}, \quad \text{where } \mathbf{p}_t = \pi_{\theta}(\cdot | \mathcal{R}_{<t}, x; T) = \text{Softmax}\left(\frac{\mathbf{z}_t}{\tau}\right). \quad (1)$$

This entropy quantifies the uncertainty in the token generation.

3.2 Entropy-based Pilot Experiments

In this section, we delve into two high-entropy challenges in Web agent training and quantify their limitations.

3.2.1 Problem-1: High-Entropy Rollout Collapse. We select ARPO as the backbone algorithm [10], a representative entropy-guided agentic RL method, and train with its default 1k training samples. We further quantify: (i) the steps in each sampled trajectory that exhibit consecutive high-entropy tool usage; (ii) within each rollout batch (branching budget is 8), the number and probability of trajectories that contain high-entropy branches.

As shown in Figure 2 (left), our key findings are: **(i) High-entropy tool-call turns exhibit transitivity:** the proportion of consecutive high-entropy tool-call turns (56.5%) exceeds isolated turns (43.5%), with trajectories reaching up to 6 consecutive high-entropy turns. This indicates that high-entropy tool-call rounds

often occur consecutively. **(ii) Rollout branch collapse:** 93.4% of branches concentrate on 1–3 trajectories, while the remaining trajectories receive virtually no budget for high-entropy branch sampling. This shows an imbalanced allocation of rollout branching resources. We argue that two observations are tightly coupled: **Due to an excessive number of consecutive high-entropy rounds in specific samples, the model tends to over-branch on a few trajectories during the rollout phase.** We define this issue as the “High-Entropy Rollout Collapse”.

3.2.2 Problem-2: High-Entropy Token Gradient Clipping. Under the same setup as previous experiment, we further visualize the policy update phase of ARPO, including (i) the importance sampling ratio of tokens in trajectories; (ii) a comparison of the Top-10 gradient-clipped tokens between ARPO and DAPO during a training step, with clipping thresholds of 0.2 and 0.28.

As illustrated in Figure 2 (right), we identify the following insights: **(i)** Consistent with findings in single-turn RL efforts [5, 38], tokens related to logical transitions, connections, and reflections typically exhibit high entropy. Beyond this, specific tool-call tokens also show high entropy. These tokens are highly functional and have low contextual dependency, incentivizing the model to explore diverse reasoning paths and tool-use patterns. **(ii)** Vanilla RL method uniformly clips the gradients of high-entropy tokens without distinguishing whether they include valuable exploratory behaviors. Although DAPO adopt clip-higher strategy [66] to alleviate this by increasing the threshold, the clipping distribution remains similar, and the clipped token count is still substantial.

Moreover, we empirically find that significant gradient clipping emerges in the very first policy update, resulting in a **lack of gradient support for high-entropy exploratory tokens in early training**. This leads to fixed paradigmatic reasoning, hindering the LLM from exploring tool-use patterns. We define this as the “High-entropy Token Gradient Clipping”.

3.3 Agentic Tool Design

In this paper, we focus on exploring entropy-balanced optimization for web agent RL algorithms. To this end, we align with existing work on web agent RL training [10, 32, 59] and select three of the most representative tools to evaluate the effectiveness of AEPO: **(1) Web Search Engine:** Provides retrieved source text and corresponding URL information from the web in response to user queries. **(2) Web Browser:** Accesses and parses URL information returned by the search engine, then summarizes the content. **(3)**

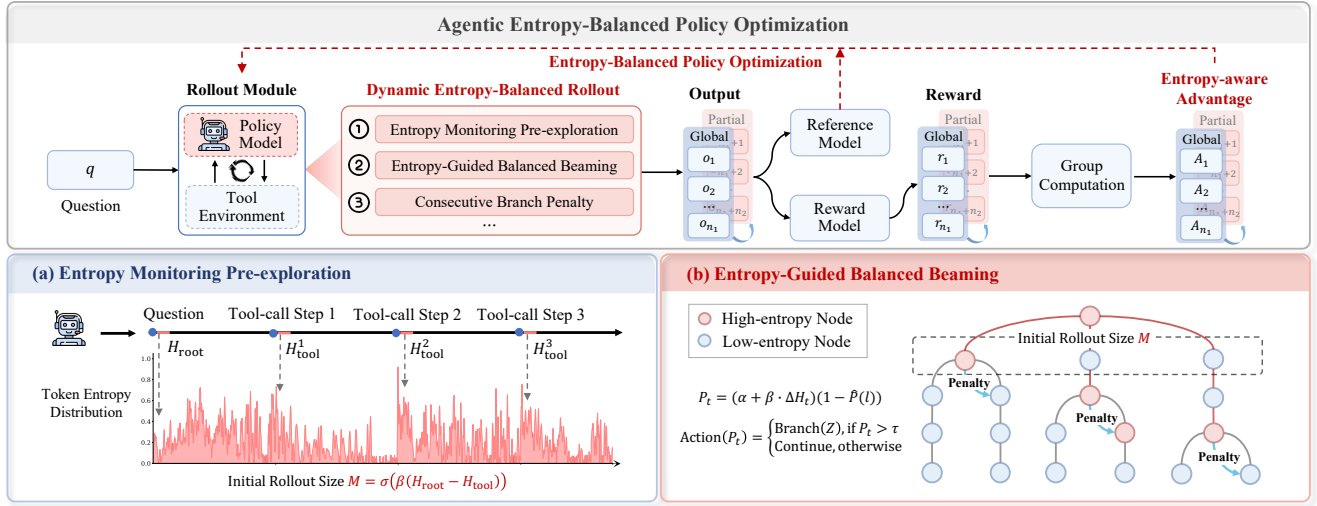


Figure 3: The overview of Agentic Entropy-Balanced Policy Optimization.

Code Executor: Executes code generated by models, returning the execution results or error messages.

4 Methodology

We introduce Agentic Entropy-Balanced Policy Optimization (AEPO), an agentic RL algorithm proposed to balance entropy during both rollout and policy update phases. As shown in Figure 3, AEPO comprises two core components:

- (1) **Dynamic Entropy-Balanced Rollout:** To mitigate the “*High-Entropy Rollout Collapse*” identified in pilot experiments (§3.2), we adaptively allocate the sampling budget between global and branch sampling via entropy pre-monitoring (§4.1.1), and penalize consecutive high-entropy tool-call steps during rollout to avoid over-branching (§4.1.2).
- (2) **Entropy-Balanced Policy Optimization:** To further address “*High-Entropy Token Gradient Clipping*”, we insert a stop-gradient operation into the clipping term to preserve and properly rescale gradients on high-entropy tokens (§4.2.1), while incorporating entropy-aware advantage estimation to prioritize learning on high-uncertainty tokens (§4.2.2).

Below, we will delve into the specifics of our approach.

4.1 Dynamic Entropy-Balanced Rollout

In this section, we address the “*High-Entropy Rollout Collapse*” by naturally breaking it down into two sub-goals: (1) Providing more reasonable resource allocation for global and branch sampling; (2) Penalizing continuous high-entropy branch sampling within single trajectories. Consequently, we propose the following two algorithmic solutions.

4.1.1 Entropy Pre-Monitoring. Traditional tree-based rollout empirically allocates resources for global and branch exploration without theoretical support [10, 36]. Inspired by the information bottleneck theory [55], we advocate allocating exploration resources from the perspective of maximizing information gain. Specifically, given a total sampling budget of k , which includes m global samples

and $k - m$ high-entropy partial branch samples, we simply model the sampling information gain I_{Gain} per rollout step as:

$$I_{\text{Gain}} = \underbrace{m \cdot I_{\text{root}}}_{\text{Global}} + \underbrace{(k - m) \cdot I_{\text{tool}}}_{\text{Partial}}. \quad (2)$$

Here, I_{root} and I_{tool} represent the information gain from the input question and external tool-call result. In the autoregressive decoding process of a language model, the information gain of the question is typically measured by the token entropy decoded by the model, with informative questions generally leading to greater uncertainty [4, 71]. Therefore, we derive the following positive correlation:

$$I_{\text{Gain}} \propto \underbrace{m \cdot H_{\text{root}}}_{\text{Global}} + \underbrace{(k - m) \cdot H_{\text{tool}}^{\text{avg}}}_{\text{Partial}}, \text{ where } H_{\text{tool}}^{\text{avg}} = \frac{1}{N} \sum_{i=1}^N H_{\text{tool}}^i. \quad (3)$$

where H_{root} and H_{tool}^i represent the entropy of the question and the entropy introduced by the i -th tool call, respectively. Based on the formula, we reveal that: (1) When $H_{\text{root}} - H_{\text{tool}}^{\text{avg}} > 0$, the uncertainty from the initial question surpasses that from the subsequent tools. In this case, we should increase m to enhance global exploration, thereby boosting the information gain I_{Gain} . (2) Conversely, when $H_{\text{root}} - H_{\text{tool}}^{\text{avg}} < 0$, m should be decreased to allocate more budget to branch exploration via tool calls.

Based on the above theoretical analysis, we propose the entropy pre-monitoring phase. As shown in Figure 3(a), we first allow the LLM to generate a complete tool-integrated trajectory for the input q . Following ARPO’s entropy calculation [10], we compute the question and tool average entropies for each trajectory according to Equation (3), forming the entropy matrices H_{root} and $H_{\text{tool}}^{\text{avg}} \in \mathbb{R}^{1 \times k}$. Subsequently, by comparing the H_{root} and $H_{\text{tool}}^{\text{avg}}$, we dynamically determine the global sampling count m as:

$$m = k \cdot \sigma \left(\beta \cdot (H_{\text{root}} - H_{\text{tool}}^{\text{avg}}) \right), \quad (4)$$

where $\sigma(x)$ is the sigmoid function, and β controls sensitivity. The value of m is positively correlated with the entropy gap between

the question and the tools. As a result, AEPO dynamically allocates rollout sampling resources, thereby enabling efficient sampling.

4.1.2 Entropy-Balanced Adaptive Rollout. After the entropy pre-monitoring stage, we introduce the main entropy-balanced adaptive rollout phase to penalize consecutive high-entropy branch sampling, which comprises three steps:

(1) **Entropy Variation Monitoring:** Following resource allocation from the pre-monitoring phase, the LLM generates m trajectory-level samples for the query q , recording the initial entropy matrix H_{root} for each trajectory. After each tool-call step t , the real-time entropy of the model's output is continuously monitored and represented as a step-level entropy matrix $H_t \in \mathbb{R}^{1 \times k}$. The standardized entropy variation relative to the initial entropy is then computed as $\Delta H_t = \text{Normalize}(H_t - H_{\text{root}})$, where the normalization involves dividing the sum of all values in ΔH by the vocabulary size V .

(2) **Entropy-Balanced Beaming:** Unlike traditional entropy-guided branch sampling [10, 70], AEPO promotes adaptive exploration that showcases beneficial entropy changes in tool-call steps while also constraining consecutive high-entropy branch sampling in specific chains. As shown in Figure 3(b), we introduce a consecutive branch penalty strategy. Given a tool-call step t , the number of consecutive high-entropy branches l prior to step t for each chain is tracked, defining the branch sampling probability at step t as:

$$P_t = (\alpha + \gamma \cdot \Delta H_t)(1 - \hat{P}(l)), \quad (5)$$

where α is the base sampling probability and γ is the entropy stabilization factor. $\hat{P}(l)$ is a linear function related to l . P_t decreases as the number of consecutive branching steps l increases, implementing a high-entropy branching penalty. We then define the rollout branching action at step t as:

$$\text{Action}(P_t) = \begin{cases} \text{Branch}(Z), & \text{if } P_t > \tau, \\ \text{Continue}, & \text{otherwise.} \end{cases} \quad (6)$$

When P_t exceeds the predetermined threshold τ , $\text{Branch}(Z)$ is initiated, creating Z partial branching reasoning paths from the current node; otherwise, the current trajectory continues.

(3) **Termination Conditions:** Our iterative rollout process terminates when one of the following conditions is met: (a) If the number of branch paths Z^* reaches the partial sampling budget $k - m$, branching stops and sampling continues until the final answer is generated; (b) If all paths terminate before reaching $k - m$, then $k - m - Z^*$ additional trajectory-level samples are added to satisfy condition (a).

Through dynamic entropy-balanced rollouts, AEPO ensures the diversity of sampling branches while adaptively allocating exploration resources, addressing the "High-entropy Rollout Collapse".

4.2 Entropy-Balanced Policy Optimization

AEPO preserves a considerable number of exploratory tokens, presenting a challenge in effectively updating these tokens' gradients. This section aims to improve targeted learning for these tokens by implementing following designs:

4.2.1 Entropy Clipping-Balanced Mechanism. Unlike traditional methods that entirely discard gradients outside the clipping range [10, 45], AEPO introduces an innovative high-entropy clipping-balanced

mechanism. The core idea is to retain high-entropy gradients that exceed the clipping interval, allowing the model to learn valuable exploratory token signals. Specifically, we integrate a *stop-gradient* operation into the high-entropy clipping term of the policy update phrase, decoupling forward and backward propagation. Our mechanism ensures that forward propagation remains unchanged, while protecting the gradient backward of high-entropy tokens from clipping constraints. For instance, in GRPO [45], given an input question x and a policy model y , GRPO enables the reference policy π_{ref} to generate a group of G outputs $\{y_1, y_2, \dots, y_G\}$ and optimizes the policy by maximizing:

$$\mathcal{L} = \mathbb{E}_{x \sim \mathcal{D}} \left[\frac{1}{\sum_{j=1}^G T_j} \sum_{j=1}^G \sum_{t=1}^{T_j} \min \left(\delta \tilde{A}^{(t)}, \text{clip} \left(\delta, 1 - \epsilon_l, \frac{1 + \epsilon_h}{\text{sg}(\delta)} \delta \right) \tilde{A}^{(t)} \right) \right], \quad (7)$$

where $\delta = r_t^{(j)}(\theta)$ represents the importance sampling ratio, and $\text{sg}(\cdot)$ denotes the *stop-gradient* operation. It is noteworthy that **the value of the term $\delta \cdot \text{sg}(\delta)$ always equals 1, ensuring that AEPO's forward computation remains unchanged.** For the backpropagation, AEPO's gradient update process is formulated as:

$$\nabla_{\theta} \mathcal{L} = \mathbb{E}_{x \sim \mathcal{D}} \left[\frac{1}{\sum_{j=1}^G T_j} \sum_{j=1}^G \sum_{t=1}^{T_j} \mathcal{F}_{j,t}(\theta) \cdot \phi_{\theta}(a_{j,t}, s_{j,t}) \cdot \tilde{A}^{(t)} \right], \quad (8)$$

where $\mathcal{F}_{j,t}(\theta) = \begin{cases} 1 + \epsilon_h, & \text{if } \delta > 1 + \epsilon_h \text{ and } \tilde{A}^{(t)} > 0, \\ 0, & \text{if } \delta < 1 - \epsilon_h \text{ and } \tilde{A}^{(t)} < 0, \\ \delta, & \text{otherwise.} \end{cases}$

During backpropagation, the gradient of a high-entropy token is retained and appropriately rescaled to $1 + \epsilon_h$ only when $\delta > 1 + \epsilon_h$ and $\tilde{A}^{(t)} > 0$. In other cases, the gradient update rule aligns with the vanilla clipping mechanism of GRPO. This controlled rescaling ensures that the model learns a balanced exploratory behavior without completely ignoring high-entropy tokens. For detailed proof of the AEPO's gradient form, please refer to the Appendix A.

4.2.2 Entropy-aware Advantage Estimation. Owing to the clipping preserve mechanism, we retain the gradients of high-entropy tokens. However, a challenge arises in training the model to better distinguish between exploratory and non-exploratory tokens. Traditional outcome-based RL algorithms assign the same advantage to all tokens in a sequence based on the answer correctness, neglecting the model's confidence levels across different tokens [22, 45].

To this end, we propose an **entropy-aware advantage estimation** that incorporates token entropy calculation into advantage shaping. This approach allows the model to assign greater rewards to exploratory tokens that are correct but exhibit high uncertainty. A natural way is to calculate an accuracy-based advantage while integrating an entropy-based advantage term, defined as follows:

$$\tilde{A}_{\text{Acc}}^{(t)} = \frac{r_t - \text{mean}(\{R_i\}_{i=1}^G)}{\text{std}(\{R_i\}_{i=1}^G)}, \quad \tilde{A}_{\Delta H}^{(t)} = \frac{H_t - \text{mean}(\{H_t\}_{t=1}^T)}{\text{std}(\{H_t\}_{t=1}^T)}, \quad (9)$$

where H_t represents the t -th token entropy, and T is the total number of tokens across all trajectories in the group. We estimate the entropy advantage for each token based on the average token entropy within the same trajectory. Furthermore, we treat the entropy advantage as a regularization term to reshape A_{acc} as:

$$\tilde{A}^{(t)} = \tilde{A}_{\text{Acc}}^{(t)} * (1 + a \cdot \tilde{A}_{\Delta H}^{(t)}). \quad (10)$$

This step is computed before the policy update. Notably, our entropy-aware advantage estimation can be seamlessly integrated with existing agentic RL algorithms to further enhance the model's emphasis on learning exploratory tokens during training.

5 Experiment Settings

Datasets. We assess the effectiveness of AEPO in web agents' RL training through three long-term reasoning tasks: (1) **Deep Information Seeking Tasks:** This includes challenging evaluations for web agents: General AI Assistant (GAIA) [39] and the Human Last Exam (HLE) [42], as well as deep information seeking: Webwalker [60], XBench [2], and Frames [28]. (2) **Knowledge-Intensive Reasoning:** This covers 3 multi-hop complex open-domain question answering tasks: 2WikiMultiHopQA [21], Musique [56], and Bamboogle [43], along with the web multi-hop task Webwalker [61]. (3) **Computational Reasoning:** This includes GSM8K [7], and competition-level math challenges: MATH [20], MATH500 [37], AIME2024, and AIME2025. All dataset splits align with the standard settings established by previous works [10, 27, 35].

Baselines and Evaluation Metrics. We consider the following strong baseline methods: (1) **Advanced RL Algorithms:** We select three categories of RL algorithms: (1) Vanilla RL: GRPO [45] and Reinforce++ [22]; (2) Clipping-optimized RL: DAPO [66], CISPO [40] and GPPO [50]; and (3) Agentic RL: GIGPO [15] and ARPO [10]. (2) **Advanced Backbone Models:** For challenging reasoning benchmarks, we evaluate the instruction-tuned versions of Qwen2.5 [44] and Llama3.1 [12]. For deep information seeking tasks, we also report results for QwQ [54], DeepSeek-R1 [18], GPT-4o [23], and o1-preview [23], using Qwen3-32B [64] as a reference. (3) **Advanced Web Agents:** We introduce a series of open-source workflow-based search agents as references, including vanilla RAG [29], Search o1 [33], Webthinker [35], and ReAct [65]. The detailed introductions of baselines are listed in Appendix C.1

Consistent with previous work, we use the F1 score to evaluate four question-answering tasks. For other tasks, we employ the Qwen2.5-72B-instruct, using LLM-as-Judge to assess the answers. In all tasks, answers are extracted from the box in the response.

6 Experiment Results

6.1 Main Result on Deep Information Seeking

To validate the effectiveness of AEPO in challenging deep web information seeking tasks, we train the Qwen3 series models combined with AEPO using 1K open-source samples and compared them with advanced web agents and RL algorithms. As shown in Table 1, we derived the following insights:

(1) **Limitations of Advanced Large Models:** Both advanced closed-source LLMs and large-parameter open-source LLMs (e.g., GPT-4o and DeepSeek-R1-671B) perform poorly in challenging deep information seeking scenarios, particularly on the GAIA (<30%) and HLE (<10%) benchmarks. This indicates that relying solely on model internal knowledge is insufficient for complex agentic search tasks.

(2) **Strong Generalization Ability of AEPO in Deep Information Seeking:** Compared to robust web agents and advanced RL algorithms, the Qwen3-8B and 14B models combined with AEPO demonstrate exceptional performance, achieving pass@1 scores of 11.2% and 47.6% on the HLE and GAIA benchmarks, respectively.

Notably, our model is trained solely on 1k samples from an open-source dataset, showcasing its efficiency in web agent training.

(3) **Importance of Dual Entropy Balancing Optimization:** AEPO consistently outperforms ARPO in both average performance and individual benchmarks, with Qwen3-8B showing a significant 6% improvement on the GAIA benchmark and 14B achieving a 5% increase on WebwalkerQA. This highlights the importance of AEPO's algorithmic design, which implements dual entropy balancing in both the Rollout and policy update phases, effectively facilitating LLMs' exploratory tool behavior.

6.2 Main Result on Generalized Reasoning

To further validate the effectiveness of AEPO in web agent training, we conduct a comparison of AEPO with 7 RL algorithms across 10 challenging reasoning tasks. Results are shown in Table 2. We find:

(1) **Instability of Clipping RL Algorithms in Web Agent Training:** Using GRPO as a baseline, clipping-optimized RL algorithms perform well on Qwen 2.5-7B-instruct, with GPPO and CISPO achieving average scores above 57%. However, in Llama3-8B, they do not show significant improvement over GRPO. Furthermore, practical experiments reveal that clipping-optimized RL algorithms often lead to entropy collapse, disrupting training performance. This indicates that clipping-optimized RL algorithms are sensitive to the architecture of the backbone model and show instability.

(2) **Generalization Ability of Agentic RL Algorithms:** Agentic RL algorithms, represented by ARPO and GIGPO, demonstrate stable and robust performance across both backbone models, with ARPO achieving average performance consistency above 55%. Notably, these methods attempt tree-structured rollout during the rollout phase, further confirming the effectiveness of branching exploration in high-entropy tool-call steps.

(3) **Effectiveness of AEPO:** AEPO consistently outperforms other reinforcement learning algorithms across 10 datasets and backbone models, achieving an accuracy improvement of nearly 5% over GRPO while maintaining competitiveness across fine-grained domains. These results highlight AEPO's efficiency and strong adaptability across different model architectures and tasks, making it more suitable for training multi-turn web agents.

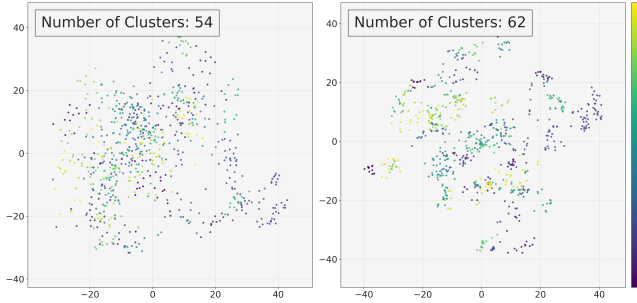
6.3 Does AEPO Mitigate Rollout Collapse?

(1) **Diversity Analysis.** To investigate whether AEPO's dynamic entropy-balanced rollout improves sampling diversity, we follow the setup of the preliminary experiment (§3.2) and randomly select samples from 10 rollout steps, encompassing 640 distinct problems and approximately 7.6k trajectories. We further employ BGEM3 [1] as the semantic embedding model, applied the PCA method for dimensionality reduction, and used DBSCAN [13] for clustering to visualize the representation of rollout sampling.

As shown in Figure 4, the results indicate that compared to ARPO, AEPO's sampling trajectories form more distinct cluster centers (54 vs. 62) and exhibit tighter intra-cluster distances with larger inter-cluster gaps. This demonstrates that AEPO improves the scope of rollout diversity and provides clearer differentiation in the sampling path distribution. We attribute this to AEPO's entropy pre-monitoring and continuous entropy penalty branches,

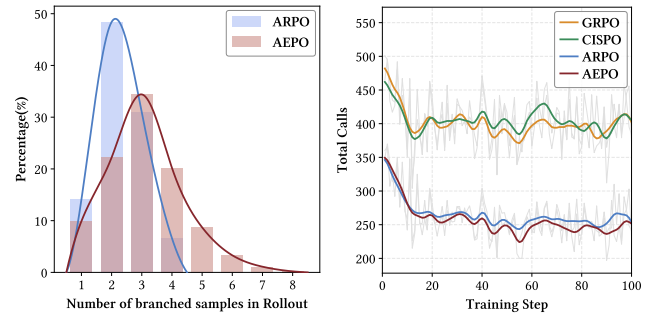
Table 1: Overall performance on deep information seeking tasks. The best results are in bold, and the second-best results are underlined. Results from larger or closed-source models are presented in gray for reference.

Method	General AI Assistant				WebWalkerQA				Humanity’s Last Exam				XBench-DR	FRAMES
	Lv.1	Lv.2	Lv.3	Avg.	Easy	Med.	Hard	Avg.	NS	CE	SF	Avg.	Avg.	Avg.
Direct Reasoning (>=32B)														
Qwen3-32B-thinking	26.2	12.1	0	14.9	6.9	1.1	2.9	3.1	<u>14.6</u>	<u>9.8</u>	8.4	12.6	14.0	26.0
DeepSeek-R1-32B	21.5	13.6	0.0	14.2	7.5	1.4	4.2	3.8	6.6	5.1	6.5	6.4	10.0	23.8
QwQ-32B	30.9	6.5	5.2	18.9	7.5	2.1	4.6	4.3	11.5	7.3	5.2	9.6	10.7	28.8
GPT-4o	23.1	15.4	8.3	17.5	6.7	6.0	4.2	5.5	2.7	1.2	3.2	2.6	18.0	44.6
DeepSeek-R1-671B	40.5	21.2	5.2	25.2	5.0	11.8	11.3	10.0	8.5	8.1	9.3	8.6	32.7	45.6
o1-preview [†]	-	-	-	-	11.9	10.4	7.9	9.9	12.9	8.1	6.6	11.1	-	-
Single-Enhanced Method (Qwen3-8B)														
Vanilla RAG	28.2	15.4	16.7	20.4	8.9	10.7	9.9	10.0	5.1	1.6	<u>12.9</u>	5.8	8.0	18.8
Search-o1	35.9	15.4	0.0	21.4	6.7	15.5	9.7	11.5	7.6	2.7	5.3	6.4	10.0	19.2
WebThinker	43.6	11.5	0.0	22.3	6.7	13.1	16.9	13.0	7.3	4.0	6.3	6.6	13.0	21.4
ReAct	35.9	17.3	<u>8.3</u>	23.3	8.9	16.7	18.3	15.5	4.2	4.0	6.3	4.6	16.0	21.1
RL-based Method (Qwen3-8B)														
Qwen3-8B	28.1	15.4	16.7	20.4	0.0	2.4	2.8	2.0	3.9	2.7	8.4	4.6	9.0	19.0
+ GRPO	48.7	25.0	<u>8.3</u>	32.0	24.4	33.3	26.8	29.0	7.9	4.0	10.5	7.8	20.0	46.2
+ ARPO	<u>53.9</u>	<u>32.7</u>	16.7	<u>38.8</u>	26.7	<u>33.3</u>	<u>29.6</u>	<u>30.5</u>	7.3	6.7	15.8	<u>8.8</u>	<u>25.0</u>	<u>47.8</u>
+ AEPO (Ours)	61.5	42.3	<u>8.3</u>	45.6	22.2	39.3	35.2	34.0	12.1	<u>5.3</u>	11.6	11.0	28.0	50.2
Single-Enhanced Method (Qwen3-14B)														
Vanilla RAG	38.5	19.2	<u>8.3</u>	25.2	17.8	13.1	11.3	13.5	5.5	6.3	9.4	6.0	15.0	31.4
Search-o1	48.7	23.1	0.0	30.1	11.1	21.4	16.9	17.5	6.4	4.0	10.5	6.8	21.0	39.8
WebThinker	48.7	26.9	<u>8.3</u>	33.0	13.3	23.8	18.3	19.5	7.0	4.0	9.5	7.0	23.0	40.8
ReAct	48.7	25.0	<u>8.3</u>	32.0	11.1	20.2	12.7	15.5	5.8	5.3	10.5	6.6	20.0	37.6
RL-based Method (Qwen3-14B)														
Qwen3-14B	33.3	13.5	0.0	19.4	6.7	2.4	4.2	4.0	5.5	6.7	11.6	6.8	14.0	23.8
+ GRPO	51.3	34.6	0.0	36.9	28.9	33.3	26.8	30.0	7.9	6.7	<u>12.6</u>	8.6	<u>27.0</u>	54.6
+ ARPO	56.4	40.4	16.7	<u>43.7</u>	<u>31.1</u>	<u>42.9</u>	<u>31.0</u>	<u>36.0</u>	<u>10.3</u>	<u>10.7</u>	13.7	<u>10.0</u>	32.0	55.4
+ AEPO (Ours)	61.5	44.2	16.7	47.6	42.2	45.2	35.2	41.0	10.6	14.7	10.5	11.2	32.0	58.8

**Figure 4: Rollout diversity: ARPO (left) and AEPO (right)**

which effectively address the continuity of high-entropy branches to achieve comprehensive coverage of the problem-solving space.

(2) Statistics Analysis. To quantitatively analyze AEPO’s effectiveness in addressing rollout collapse, we measure the branch distribution of ARPO and AEPO over 10 steps during rollout. As shown in Figure 5 (left), with a global sampling batch size of 8, ARPO typically branches into 2-3 trajectories. In contrast, AEPO exhibits a more diverse branching pattern, potentially covering all 8 paths with different branches. This highlights AEPO’s dynamic resource allocation and continuous branch penalty mechanism enable the model to explore high-entropy tool-call steps across different trajectories, effectively mitigating bias in specific path branches.

**Figure 5: Branch sampling distribution in rollout (left), and tool-call efficiency across four RL algorithms (right).**

6.4 Does AEPO Achieve Entropy-Balanced and Efficient RL Training?

(1) Tool-call Efficiency Analysis. In agentic RL training, effectively controlling the frequency of tool usage can significantly reduce financial costs. To confirm the efficiency of AEPO’s tool usage, we quantify the tool consumption of AEPO compared to other RL algorithms. As shown in Figure 5 (right), AEPO requires only about half the number of tool calls to achieve superior performance compared to vanilla and clipping-optimized RL algorithms. Additionally, compared to the agentic RL algorithm ARPO, AEPO consistently reduces the number of tool calls. We attribute this

Table 2: Overall performance on ten challenging reasoning tasks. The top two outcomes are bolded and underlined.

Method	Mathematical Reasoning					Knowledge-Intensive Reasoning					Avg.
	AIME24	AIME25	MATH500	GSM8K	MATH	WebWalker	HotpotQA	2Wiki	MuSiQue	Bamboogle	
Backbone Model: Llama3.1-8B-Instruct											
Classical RL Method											
+ GRPO	13.3	<u>13.3</u>	62.4	87.4	79.2	26.5	57.8	71.8	31.0	68.2	51.1
+ Reinforce ++	13.3	16.7	61.4	87.0	77.2	27.5	57.1	71.6	29.9	69.1	51.1
Clipping-optimized RL Method											
+ DAPO	16.7	<u>13.3</u>	61.2	87.4	76.4	25.5	56.6	70.3	29.2	67.3	50.4
+ GPPO	16.7	<u>6.7</u>	61.8	86.6	79.4	27.5	61.8	72.8	29.8	71.9	51.5
+ CISPO	13.3	6.7	62.2	87.0	78.2	26.0	57.3	<u>75.6</u>	32.2	71.8	51.0
Agentic RL Method											
+ GIGPO	20.0	13.3	62.4	87.4	77.2	<u>31.5</u>	61.8	74.6	31.8	72.1	53.2
+ ARPO	23.3	16.7	64.6	88.0	80.2	30.5	65.4	75.5	34.8	73.8	55.3
+ AEPO (Ours)	26.7	16.7	65.8	<u>87.6</u>	80.6	33.5	<u>64.7</u>	79.0	<u>33.0</u>	75.8	56.3
Backbone Model: Qwen2.5-7B-Instruct											
Classical RL Method											
+ GRPO	23.3	<u>26.7</u>	78.0	92.8	87.8	22.0	59.0	<u>76.1</u>	30.6	68.4	56.5
+ Reinforce ++	26.7	23.3	78.0	<u>92.2</u>	<u>88.8</u>	26.0	55.1	68.9	25.2	64.9	54.9
Clipping-optimized RL Method											
+ DAPO	20.0	23.3	80.4	91.0	<u>88.8</u>	24.0	57.7	68.4	28.6	65.5	54.8
+ GPPO	26.7	23.3	76.2	91.6	87.6	<u>31.0</u>	<u>60.7</u>	74.2	31.5	<u>72.4</u>	57.5
+ CISPO	26.7	30.0	77.8	91.4	86.2	29.0	59.3	72.1	29.1	70.2	57.2
Agentic RL Method											
+ GIGPO	<u>30.0</u>	20.0	78.4	91.6	87.6	30.5	58.1	73.5	<u>31.1</u>	70.1	57.1
+ ARPO	<u>30.0</u>	30.0	<u>78.8</u>	<u>92.2</u>	<u>88.8</u>	26.0	58.8	<u>76.1</u>	<u>31.1</u>	71.5	<u>58.3</u>
+ AEPO (Ours)	33.3	30.0	80.4	<u>92.2</u>	90	31.5	62.5	77.1	<u>31.1</u>	73.4	60.1

enhanced efficiency to the entropy pre-monitoring phase, which balances the allocation of rollout exploration resources based on the information gain from the problem and tool usage. This ensures that AEPO not only broadens the rollout exploration space but also achieves efficient web agent training.

(2) *Entropy Stability Analysis.* To better quantify the effectiveness of entropy-balanced policy optimization during policy updates, we present the RL training curves for 10 reasoning tasks. Figure 6 illustrates the dynamic visualization of entropy loss and validation set accuracy across 10 benchmark tasks throughout the training steps. We observe that using clipping-optimized RL often encounters entropy instability during training, leading to performance collapse. In contrast, AEPO demonstrates a more stable entropy curve compared to other reinforcement learning algorithms. Notably, sudden increases or decreases in entropy loss do not enhance training stability. Instead, maintaining relatively high and stable entropy is generally beneficial for sustained performance improvement. This finding aligns with our motivation, as AEPO introduces entropy-balanced policy optimizations to shape more reasonable and stable entropy dynamics.

7 Conclusion

In this paper, We propose Agentic Entropy-Balanced Policy Optimization (AEPO), an agentic RL algorithm that balances entropy in both rollouts and policy updates. We first identify two entropy-induced challenges via preliminary experiments. AEPO has two key components: (1) a dynamic entropy-balanced rollout that pre-monitors entropy to allocate budget between global and branch

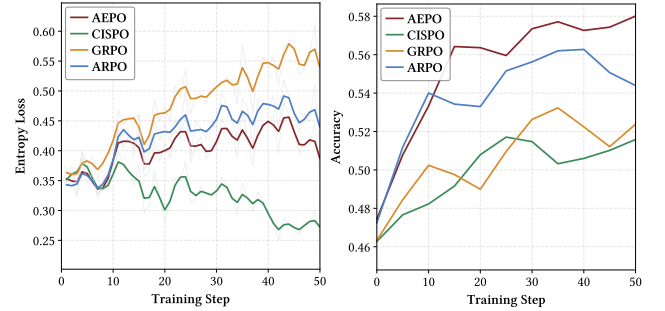


Figure 6: Visualization of training dynamics, including entropy loss(left) and accuracy (right) across training steps

sampling, and applies a penalty to consecutive high-entropy tool-call steps to avoid oversampling; (2) an entropy-balanced policy optimization that uses a stop-gradient high-entropy clipping term to preserve and rescale gradients on high-entropy tokens, together with entropy-aware advantage estimation to prioritize learning on high-uncertainty tokens. Across 14 benchmarks, AEPO consistently outperforms seven mainstream agentic RL algorithms. Further analyses verify its scalability and stability, offering practical guidance for training general web agents.

Acknowledgments

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Appendix

A Proof of the Gradient of AEPO

In this section, we will comprehensively detail the theoretical derivation of AEPO’s forward propagation formulas and how they lead to the backward propagation formulas. Specifically:

We begin with the loss function:

$$\mathcal{L} = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \left[\frac{1}{\sum_{j=1}^G T_j} \sum_{j=1}^G \sum_{t=1}^{T_j} \min \left(\delta \tilde{A}^{(t)}, \text{clip} \left(\delta, 1 - \epsilon_l, \frac{1 + \epsilon_h}{\text{sg}(\delta)} \delta \right) \tilde{A}^{(t)} \right) \right], \quad (11)$$

where $\delta = r_t^{(j)}(\theta)$ represents the importance ratio, and $\text{sg}(\cdot)$ is the stop-gradient operator. Given that $\tilde{A}^{(t)}$ is a constant and $\nabla_{\theta} \delta = \delta \phi_{\theta}(a_{j,t}, s_{j,t})$, the gradient of $f(\delta)$ can be expressed as:

$$\nabla_{\theta} f(\delta) = \tilde{A}^{(t)} s(\delta) \delta \phi_{\theta}(a_{j,t}, s_{j,t}), \quad (12)$$

where $s(\delta)$ depends on the range of δ . Therefore, we consider three scenarios:

- (1) If $\tilde{A}^{(t)} > 0$ and $\delta > 1 + \epsilon_h$: The upper clipping boundary is active, so $\partial f / \partial \delta = (1 + \epsilon_h) / \text{sg}(\delta)$, effectively simplifying to $(1 + \epsilon_h)$.
- (2) If $\tilde{A}^{(t)} < 0$ and $\delta < 1 - \epsilon_l$: The lower clipping boundary dominates, leading to $\partial f / \partial \delta = 0$, causing the gradient to vanish.

The region is unclipped, resulting in $\partial f / \partial \delta = \delta$.

By combining all cases, we derive:

$$\mathcal{F}_{j,t}(\theta) = \begin{cases} 1 + \epsilon_h, & \tilde{A}^{(t)} > 0, \delta > 1 + \epsilon_h, \\ 0, & \tilde{A}^{(t)} < 0, \delta < 1 - \epsilon_l, \\ \delta, & \text{otherwise.} \end{cases} \quad (13)$$

Thus, the gradient update is given by:

$$\nabla_{\theta} \mathcal{L} = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}} \left[\frac{1}{\sum_{j=1}^G T_j} \sum_{j=1}^G \sum_{t=1}^{T_j} \mathcal{F}_{j,t}(\theta) \cdot \phi_{\theta}(a_{j,t}, s_{j,t}) \cdot \tilde{A}^{(t)} \right]. \quad (14)$$

B Baselines

In this section, we provide a detailed overview of the baseline models involved in all experiments, as follows:

B.1 RL algorithms

(1) Classical RL Method:

- **GRPO** [45] is a reinforcement learning algorithm for fine-tuning large language models via group-based policy optimization. It optimizes model behaviors by comparing responses within sampled groups and assigning relative rewards, enabling more stable and sample-efficient policy updates.
- **Reinforce++** [22] extends the classic policy-gradient algorithm by incorporating variance reduction and adaptive normalization techniques. It improves training stability and sample efficiency when fine-tuning language models with scalar rewards, while keeping the overall objective aligned with standard REINFORCE.

(2) Clipping-optimized RL Method

- **DAPO** [66] decouples the clipping operation from the policy update to achieve more stable optimization, and introduces a

dynamic sampling strategy that adaptively selects training examples to maintain effective gradient signals. These techniques together improve training efficiency and prevent performance degradation in long-horizon reasoning tasks.

- **GPPO** [50] extends the PPO framework by decoupling the clipping operation between the forward and backward passes. During optimization, the policy ratio is clipped in the forward computation to ensure bounded updates, while the original, unclipped ratio is used in the backward path to preserve complete gradient information.
- **CISPO** [40] reformulates ratio clipping by applying the constraint to importance sampling weights instead of policy ratios. It bounds update magnitudes in expectation while preserving token-level gradient information through unclipped policy ratios.

(3) Agentic RL Method

- **GIGPO** [15] groups complete trajectories at the episode level to compute macro-relative advantages, and also retroactively group actions sharing anchor states across trajectories at the step level to compute micro-relative advantages. Both levels are combined without using a critic, preserving the critic-free nature while enabling per-step credit signals.
- **ARPO** [10] is an RL method tailored for multi-turn LLM agents. It introduces an entropy-based adaptive rollout scheme that increases sampling in steps with high uncertainty, and incorporates an advantage attribution mechanism to assign credit across branching tool-use interactions.

B.2 Web Search Agent

- **RAG** [29] (Retrieval-Augmented Generation) combines information retrieval with generative modeling to enhance the accuracy, reliability, and timeliness of outputs. It retrieves relevant information from an external knowledge base before generating responses, addressing internal knowledge gaps, and reducing hallucinations.
- **Search-o1** [33] is a framework designed to enhance reasoning by integrating agentic RAG mechanisms with a Reason-in-Documents module. It improves accuracy, coherence, and reliability in reasoning tasks, outperforming native reasoning and traditional RAG methods in complex scenarios.
- **WebThinker** [35] is an open-source framework developed by Renmin University of China, enabling LRMs to autonomously search, explore web pages, and generate research reports. It employs direct preference optimization and iterative synthesis tools to enhance tool utilization capabilities.
- **ReAct** [65] combines reasoning and action to tackle complex tasks effectively. It allows models to generate reasoning steps and use external tools, such as search engines and databases, during decision-making, optimizing results through iterative processes.

C Supplemented Experiments

C.1 Implementation Details

We implement the AEPO algorithm using the VERL framework [46], excluding tool-call results from the loss calculation to avoid bias. Our setup includes a training batch size of 128, a PPO mini-batch size of 16, and a context length of 20K. For AEPO rollout, the global

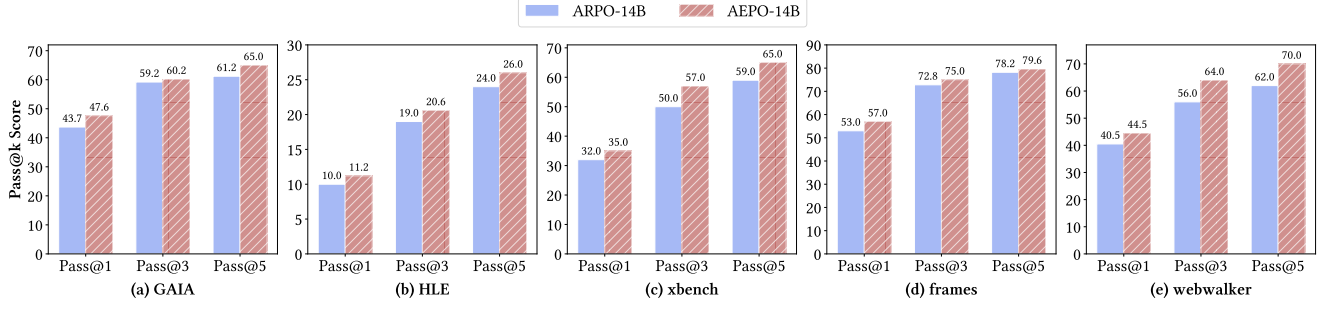


Figure 7: The comparison analysis of Qwen3-14B using ARPO and AEPO across Pass@1 to Pass@5 metrics.

rollout size is 16, with β set to 0.2. Resource allocation follows Equation 4, with a consecutive branch penalty probability of $P(l) = 0.2 \cdot l$. Other settings align with ARPO for fair comparison. To stabilize RL training, the KL divergence coefficient in GRPO is set to 0. The RL for reasoning and deep information seeking is 2 and 5 epochs. All experiments use 16 NVIDIA H800 GPUs.

During training and evaluation, we use the Bing Search API (US-EN region) as the search engine. Following RAG-related work [27, 35], we retrieve 10 web pages per query. For reasoning tasks, we use the top 10 snippets; for deep information seeking, we extract up to 6000 tokens per page and use a same size model as a browser.

C.2 Pass@K Sampling Analysis

Due to the dynamic multi-turn interactions and complexity of tool environments in web agent training, we conduct a sampling analysis of the model’s Pass@3 and Pass@5 to accurately assess its true problem-solving abilities.

As shown in Figure 7, AEPO demonstrates significant performance improvements with larger-scale sampling. Notably, the Qwen3-14B model combined with AEPO achieves remarkable results: GAIA at 65%, HLE at 26%, and xbench-DR at 62%. Compared to the robust agentic RL algorithm ARPO, AEPO consistently excels across five datasets. This stable improvement in Pass@K can be primarily attributed to AEPO’s entropy balancing optimizations, which allow the model to explore fine-grained tool usage behaviors more efficiently, thereby enhancing reasoning and sampling efficiency.